



Ivan Shuba

Machine vision PHEV charging pipeline with electricity price validation

Metropolia Ammattikorkeakoulu

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Project report

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Project idea

My current electricity contract contains basic price and flexible adjustment rates based on difference between average electricity price for month and average consumption electricity price. In another words I should try to buy energy at lowest prices or at least below day mean value.

Idea of project to make a data processing pipeline for automatic car charge with following set of rules:

- System should understand that car at home
- System should understand that charger is plugged into car
- Car should be charged before 7 am next day.
- The lowest possible price should be used to minimize electricity bills
- Forced charge option to charge car “now”

Hardware:

Hardware for this project:

1. a PHEV car which can be charged for about 6 hours,
2. Home Assistant server on Raspberry Pi,
3. Ring doorbell with camera
4. ZigBee relay switch with power meter on charger powerline.

Implementation

This pipeline was made with Python script and Homa Assistant YAML automation scripts. Ring doorbell snapshots are taken as an input data by MQTT subscription from Python code. Region of interest was defined and YOLOv8 Nano model used for car class objects detection.

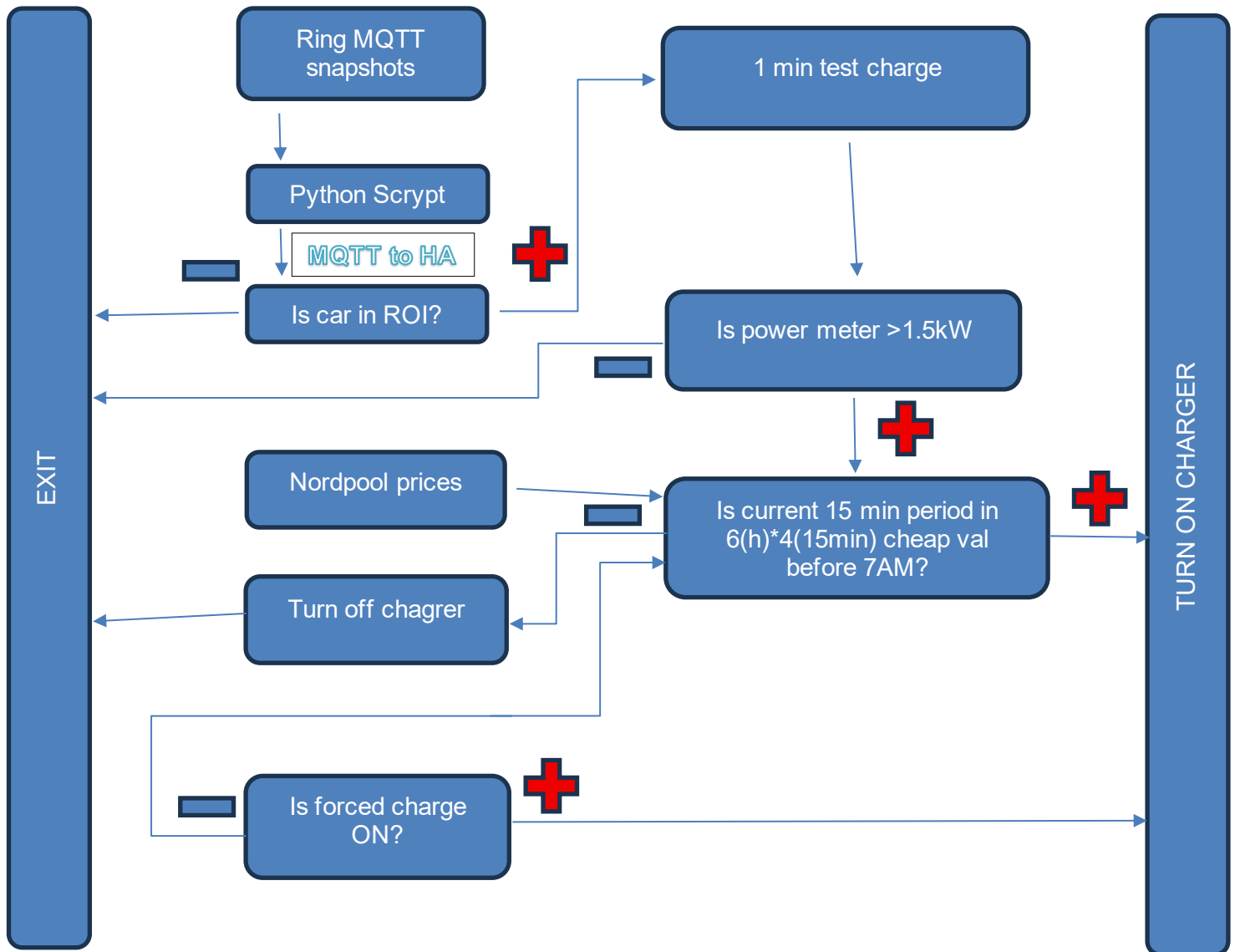


On image above default snapshot on left side and postprocessed image on the right side. Also another car and covering ROI with other objects.



Logic diagram shows pipeline logic and main control loops. Python script updates car state and push it via MQTT topic to Home Assistant. Once car detected triggered test charge script to check is charger plugged in vehicle. After 1 min of test charge instant power consumption should be over 1.5kW, then switch turns off and prices for the rest of the day and tomorrow morning are fetched from Nordpool integration. Cheapest values from this array selected. Every 15 min starting from round hour if current period price in this subarray charger switch turns on for a 15 min.

If I would like to charge car “right now” forced charge option made in automation logic. In this case charger switch turns on without checking energy prices or charging plug condition. After 6 hours it is automatically switches off to normal mode.



Ring MQTT topic freeze for 3 minutes once detects motion. But it is even better, because during that time car can be parked and plugged in. This protects system from wrong triggers and 3 minutes delay is not critical in this kind of automation.

Key Boolean values, Nordpool prices and power meter values on diagram below. As we can see first of all happens detection of car, then triggers cable connection and switch turns on at cheap time. Power meter shows that car charged before 7AM.



Noticed issues:

Due to typical PHEV position extra fitting of model with car data was not done. However, it reacts on neighbor's cars on ROI. Possible way to solve this issue: adjust ROI area or make fine tune and teach it to detect only white car.



Second improvement can be done in cheap window selection on day change. At 0:00 "today values" became "yesterday data" and automation forgets about done charging periods, because Nordpool

integration has in properties only today and tomorrow values. At 0:00 automation try to find again 6hour of cheap prices to charge car. To fix this counter can be used to look $COUNTER(MAX) - COUNTER(val)$ 15 minutes periods for charge. And once switch turns on $COUNTER(val)$ will increase by 1.

Summary:

In general, pipeline works well: PHEV car detects with a good score, automations and scripts do not confront each other.

There are couple improvements to be done like fit model on car data, cheap window calculation, notifications on not connected cable and voice control. Long-term test is required for better review and troubleshooting.